

Practical 2: ER Diagram

Aim: To study and draw an Entity Relationship (ER) diagram for an Employee - Department database scenario.

Code

This practical is conceptual (ER modeling) and does not involve SQL code. The ER model is described below.

Entity	Attributes	Primary key
DEPARTMENT	Dno, Dname	Dno
EMPLOYEE	Eno, Fname, Lname, Jdate	Eno

Relationship: WORKS_IN between DEPARTMENT and EMPLOYEE, cardinality **1 : N** (one department has many employees; each employee works in exactly one department). EMPLOYEE has **total** participation; DEPARTMENT has **partial** participation.

Output

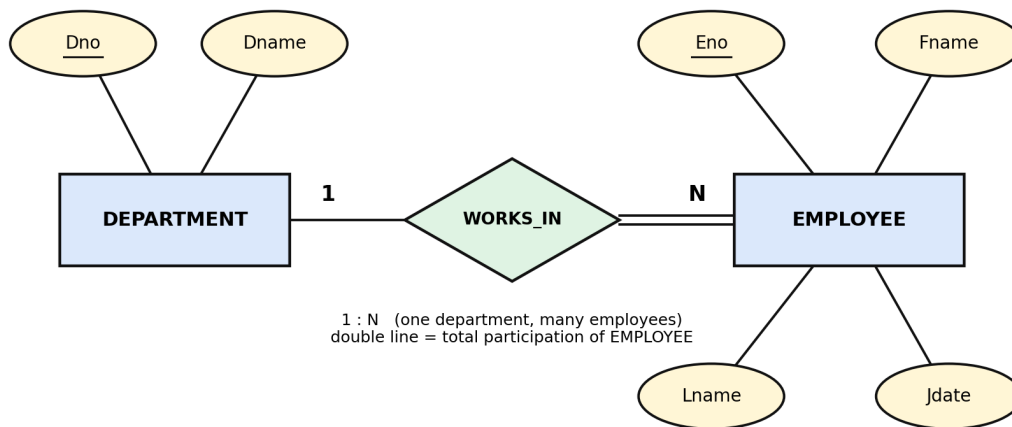


Figure 1: ER diagram (Chen notation) - DEPARTMENT WORKS_IN EMPLOYEE, 1:N. Underlined attribute = primary key; double line = total participation.

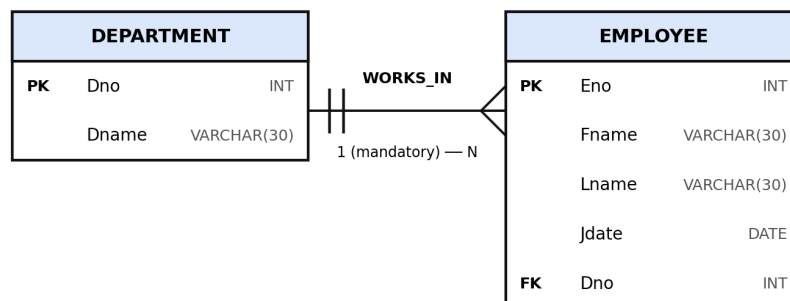
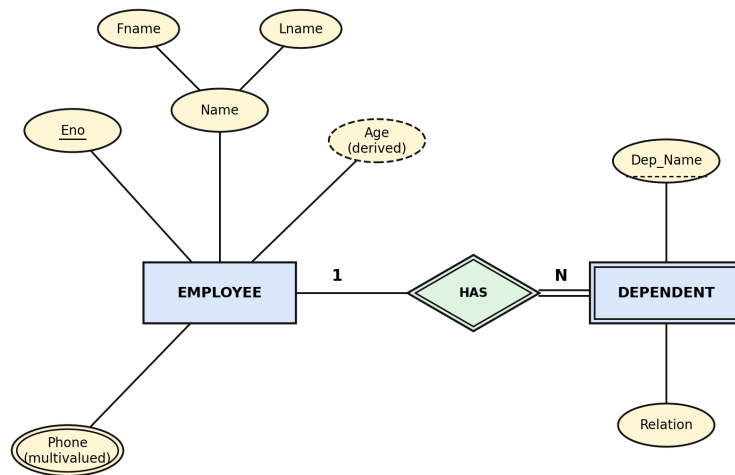


Figure 2: The same ER model in crow's-foot notation, with the foreign key Dno in EMPLOYEE.

Additional ER Diagram Concepts



Double rectangle / double diamond = weak entity and identifying relationship; double ellipse = multivalued; dashed ellipse = derived

Figure 3: Weak entity (DEPENDENT), identifying relationship (HAS), composite attribute (Name), multivalued attribute (Phone) and derived attribute (Age).

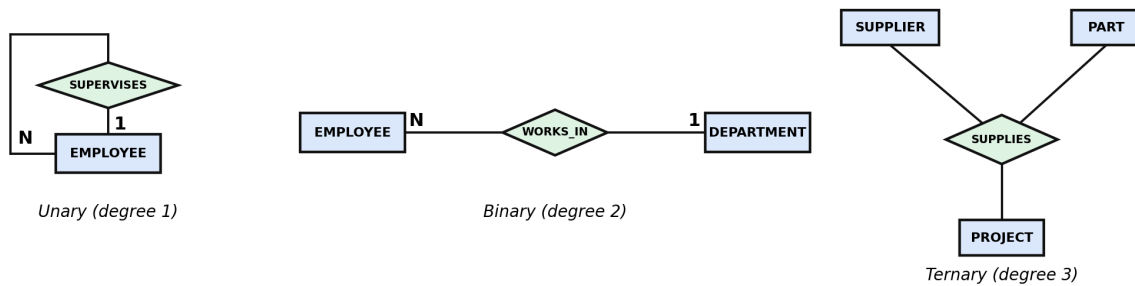


Figure 4: Degree of a relationship - unary, binary and ternary.

Result

The practical was performed successfully. The ER diagram for the DEPARTMENT - EMPLOYEE database was drawn and the desired output was obtained.